LESSON 2.1

Understand Statistical Questions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can tell the difference between a statistical question and a non-statistical question.

WHY THIS MATTERS You've just been hired as the data analyst for a youth basketball league. The head coach hands you a list of questions the staff wants to investigate about their players. Some questions will give you interesting data with variety, while others have just one fixed answer. Your job: figure out which questions are worth investigating with data. Two of the coach's questions are: "How tall is our point guard?" and "How tall is each player on the roster?"

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Statistical Question *Spanish: Pregunta estadística*	A question answered with data that are expected to vary.	How tall is each student?
Non-statistical Question *Spanish: Pregunta no estadística*	A question that does not anticipate variability in the data used to answer it. It may ask for one fact or measurement.	How tall is the flagpole?
Data *Spanish: Datos*	Facts and numbers you collect, like answers to a survey.	0, 2, 5, 1 pets
Variability *Spanish: Variabilidad*	How spread out the numbers are.	0, 2, 5 pets — answers vary
Survey *Spanish: Encuesta*	Collecting facts by asking people questions.	Asking classmates one question
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	A distribution is symmetric when its left and right sides have approximately matching shapes around the center.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Which is statistical: “How many siblings does each student have?” or “How many sides does a triangle have?”

1. Sibling counts differ: 0, 1, 3.
2. A triangle always has 3 sides.

Answer: The sibling question is statistical; the triangle question is not.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New question: "How many pets does each student in our class have?" Will every student give the same answer, or different answers?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- This question _____ statistical because _____ .
- I expect the answers to _____ from person to person.
- A non-statistical question has _____ answer.
- The data I would collect might look like _____ .

Word bank statistical | non-statistical | variability | data | survey | vary | answers | collect | fixed | different | question | typical

WATCH OUT!

A common mistake

A common mistake in Statistical Questions and Data is judging a question by its topic or its numbers instead of its variability. Students see a question about ONE player, ONE game, or ONE fixed fact

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Level 0 · Most support · every hint printed

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each question to its type.

_____	How tall is each swimmer on the team?	A	Statistical
_____	What color is the team mascot?	B	Not Statistical
_____	How many laps does each runner complete?	C	Statistical
_____	How many players are on a soccer team?	D	Not Statistical

STEPS TO TRY

1. Read each question on the left and underline the word 'each' when you see it — it's a clue that many people are being measured.
2. Sort by asking: does this question collect data from many individuals (varies), or state one fact with a single answer (doesn't vary)?
3. Start with 'What color is the team mascot?' — one mascot, one color, so match it to Not Statistical. Do the rest one question at a time.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which of the following is a statistical question?

(A) How many points did each player score this season?

(B) How many points are scored for a free throw?

(C) What sport does this team play?

(D) How many halves are in a soccer game?

STEPS TO TRY

1. Re-read each question and ask: would different people or games give different answers, or is there just one fixed answer?
2. A statistical question expects variability — many answers that differ. Sort each choice into 'one fixed answer' or 'answers that vary'.
3. Test one: 'How many halves are in a soccer game?' Every game has the same number — no variability there. Check the other three the same way.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which question is NOT a statistical question?

(A) How many home runs did each player hit this year?

(B) How many miles does each runner train per week?

(C) How many innings are in a baseball game?

(D) How tall is each player on the basketball team?

STEPS TO TRY

1. Careful — this asks for the question that is NOT statistical: the one with a single fixed answer instead of varying data.
2. For each choice ask: 'Would each player or runner give a different number?' The odd one out is a rule of the game, not data you collect.
3. Start with 'How tall is each player?' — heights differ player to player, so that one IS statistical. Keep testing until one has just one answer.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Why is 'How many hours of sleep did each student get last night?' a statistical question?

(A) Because different students got different amounts of sleep

(B) Because it asks about sleep

(C) Because hours are numbers

(D) Because the answer is always 8 hours

STEPS TO TRY

1. The question asks WHY it's statistical — focus on what happens when many students answer, not on the topic of sleep.
2. Think: if you asked every student how long they slept, would you get one number or many different numbers? That idea is called variability.
3. Picture asking 5 classmates their sleep hours and listing the numbers. Notice whether the list is all the same or varied — that's the key idea.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you notice about statistical questions compared to non-statistical questions?

SENTENCE STARTER Statistical questions expect ___ in the answers because ___. Non-statistical questions have only ___ answer because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help